Unit: 6 Unit 4: Absolute Value **Name:** ______________
Topic: Absolute Value Function and its Graph **Date:** __________

Foundation (Jalapeno)

Q1. The graph of $y = |x - 2|$ has a vertex at
(A) (0, 2)
(B) (2, 0)
(C) (2, 2)
(D) (0, 0)
(E) (1, 1)

Q2. The axis of symmetry for $y = |x + 1|$ is
(A) $x = -1$
(B) $x = 0$
(C) $x = 1$
(D) $y = -1$
(E) $y = 0$

Q3. A basketball player scores $|x - 5|$ points in a game. If x is the number of shots made, what is the minimum number of points scored?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 5

Q4. The graph of $y = 2|x|$ opens
(A) upward
(B) downward
(C) to the left
(D) to the right
(E) not at all

Q5. A tennis player runs $|x - 10|$ meters during a match. If x is the number of sets played, what is the minimum distance run?
(A) 0 meters
(B) 5 meters
(C) 10 meters
(D) 15 meters
(E) 20 meters

Q6. The graph of $y = |x| - 2$ has a vertex at
(A) (0, -2)
(B) (0, 0)
(C) (0, 2)
(D) (1, 1)
(E) (-1, 1)

Q7. A golfer's score is $|x - 3|$ strokes above par. If x is the number of holes played, what is the minimum score above par?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Q8. The graph of $y = -|x + 2|$ opens
(A) upward
(B) downward
(C) to the left
(D) to the right
(E) not at all

Open Ended Questions (FRQ)

Q9. Graph the function $y = 2|x - 1|$ and identify the vertex, axis of symmetry, and direction of opening. Show all work and explain your reasoning.

Q10. A football player's rushing yards are modeled by the function $y = |x - 5| + 2$, where x is the number of games played. Graph the function and explain the meaning of the vertex, axis of symmetry, and direction of opening in the context of the problem. Show all work and provide a clear explanation.

Chef's Tips

- Tip 1: Encourage students to use graph paper to visualize the functions.
- Tip 2: Remind students to label the vertex, axis of symmetry, and direction of opening on their graphs.
- Tip 3: Suggest that students use real-world sports scenarios to illustrate the applications of absolute value functions.